# Data Communication & Computer Networks (CN)

## Topic 3 of 7 : Addressing (MAC, IPv4, IPv6, Port, DNS)

**108 Pending Questions** — staged from repo exam PDFs & trusted sources — **NEED VERIFICATION** (answers + Hindi to be verified during topic build)

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)**

Total Questions: **108** pending questions (staging)

Question Type: MCQ — 4 options (A, B, C, D) per question, in English + Hindi

Layout: Left = **question** | Right = **explanation** (green): ✔ **why the correct answer is right** + ★ **Nukta** (key exam point)

Language rule: Hindi is the second language; **technical terms stay in English** (topology, protocol, jitter…)

Verification: Every answer cross-verified **4×** (Forouzan + official answer keys of Subject Test / Unit Test Solutions). Wrong options verified internally.

## Syllabus Coverage — sub-topic wise order

**How to read:** Left side: question + 4 options (try it yourself first). Right side (green): ✔ correct answer with why it is right, and ★ **Nukta** — the one-line exam tip to remember. Practice in 2 passes: first read question + options only, then check the green column.

**System:** generated by **tools/cn_build_topic.py** from **tools/data/topic3_data.py**. To add more questions later: drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions auto-classify topic-wise. See **Computer Network/README.md**.

## Pending Review — auto-ingested, needs verification

Auto-extracted & classified; answers to be verified before counting.

**P1.** Which of the following port numbers belong to the “well-known” ports category? "well-known"
**(A)** 80, 443, 8080
**(B)** 49152, 110, 53
**(C)** 22, 25, 80
**(D)** 1024, 3306, 8080

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ auto-ingested — answer needs verification

| | |
|---|---|
| **P2.** IP address 127.0.0.1 refers to which function? IP 127.0.0.1 function<br>**(A)** Broadcast<br>**(B)** Class D network<br>**(C)** Loopback testing<br>**(D)** APIPA<br>[Source: BCI Subject Test-8.pdf \| auto-classified: Topic 3 / IPv4 Addressing] | ⚠ auto-ingested — answer needs verification |
| **P3.** A host has IP 192.168.10.45 /27. Which statement is TRUE? Host IP 192.168.10.45<br>**(A)** Broadcast address = 192.168.10.63<br>**(B)** Network ID = 192.168.10.32<br>**(C)** First usable host = 192.168.10.33<br>**(D)** All of the above<br>[Source: BCI Subject Test-8.pdf \| auto-classified: Topic 3 / MAC Address] | ⚠ auto-ingested — answer needs verification |
| **P4.** A subnet /30 is MOST commonly used for: Subnet<br>**(A)** Two host networks<br>**(B)** Four host networks<br>**(C)** Loopback only<br>**(D)** VPN networks<br>[Source: BCI Subject Test-8.pdf \| auto-classified: Topic 3 / Subnetting & CIDR] | ⚠ auto-ingested — answer needs verification |
| **P5.** Which of the following is a class D address? - Class D address<br>**(A)** 224.1.1.1<br>**(B)** 233.0.0.10<br>**(C)** 239.5.5.5<br>**(D)** All of the above<br>[Source: BCI Subject Test-8.pdf \| auto-classified: Topic 3 / IPv4 Addressing] | ⚠ auto-ingested — answer needs verification |
| **P6.** Which range is typically used for ephemeral (dynamic) ports? ephemeral (dynamic) ports<br>**(A)** 0–1023<br>**(B)** 1024–49151<br>**(C)** 49152–65535<br>**(D)** 100–200<br>[Source: BCI Subject Test-8.pdf \| auto-classified: Topic 3 / Port Numbers & DNS/URL] | ⚠ auto-ingested — answer needs verification |
| **P7.** Which of the following port and IP address scanner is famous among users? # % B - 9 IP S „ 9 - †<br>**(A)** Cain and Abel<br>**(B)** Angry IP Scanner<br>**(C)** Snort<br>**(D)** Ettercap<br>[Source: BCI Subject Test-9.pdf \| auto-classified: Topic 3 / IPv4 Addressing] | ⚠ auto-ingested — answer needs verification |

**P8.** Which of the following uniquely identifies a process on a host? # B “ ”

**(A)** IP Address only
**(B)** Port Number only
**(C)** IP Address + Port Number
**(D)** MAC Address

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P9.** Transport layer uses which addressing? E A - E

**(A)** IP address
**(B)** MAC address
**(C)** Port number
**(D)** Domain name

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ auto-ingested — answer needs verification

**P10.** DNS works on which layer? DNS

**(A)** Transport
**(B)** Application
**(C)** Network
**(D)** Data link

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ auto-ingested — answer needs verification

**P11.** What is the usual length of a MAC address? MAC E k R

**(A)** 16 bits
**(B)** 32 bits
**(C)** 48 bits
**(D)** 64 bits

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / MAC Address]

⚠ auto-ingested — answer needs verification

**P12.** The first octet of a Class A IP address ranges from: Class A IP E W m 7 R

**(A)** 0-127
**(B)** 128-191
**(C)** 192-223
**(D)** 224-239

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P13.** How many bits are used for the network portion in a Class C IP address? Class C IP E 7 A O

**(A)** 8
**(B)** 16
**(C)** 24
**(D)** 32

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P14.** Which class of IP address is used for multicasting? f :n o IP E

**(A)** Class A
**(B)** Class B
**(C)** Class C
**(D)** Class D

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P15.** Which class of address is reserved for experimental purposes? IP E o h T 7 p q r

**(A)** Class C
**(B)** Class B
**(C)** Class D
**(D)** Class E

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P16.** Which address identifies a unique device on a local network segment? s A 7 7 E

**(A)** IP address
**(B)** MAC address
**(C)** Gateway
**(D)** Subnet Mask

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P17.** Which is a static hardware address? :s A E ?**

**(A)** IP address
**(B)** MAC address
**(C)** Port number
**(D)** Domain name

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / MAC Address]

⚠ auto-ingested — answer needs verification

**P18.** MAC address is also known as: MAC E ?**

**(A)** Logical address
**(B)** Hardware address
**(C)** IPv4 address
**(D)** Routing address

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / MAC Address]

⚠ auto-ingested — answer needs verification

**P19.** IP address is divided into how many parts? IP E 7 ?**

**(A)** 2
**(B)** 3
**(C)** 4
**(D)** 5

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P20.** What is the default subnet mask for Class B network?Class B

**(A)** 255.0.0.0
**(B)** 255.255.0.0
**(C)** 255.255.255.0
**(D)** 255.255.255.255

[Source: CN Topic Test 6.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P21.** Which IP class is reserved for multicasting? IP

**(A)** Class A
**(B)** Class B
**(C)** Class D
**(D)** Class E

[Source: CN Topic Test 6.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P22.** In IPv6, how many bits are in the address?IPv6

**(A)** 32
**(B)** 64
**(C)** 128
**(D)** 256

[Source: CN Topic Test 6.pdf | auto-classified: Topic 3 / IPv6 Addressing]

⚠ auto-ingested — answer needs verification

**P23.** Which of these DNS record types maps a domain name to an IP address? - DNS IP

**(A)** CNAME
**(B)** A
**(C)** MX
**(D)** PTR

[Source: CN Topic Test 6.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ auto-ingested — answer needs verification

**P24.** Which part of IP address defines the network in Class C?Class C IP

**(A)** First octet
**(B)** First two octets
**(C)** First three octets
**(D)** All four octets

[Source: CN Topic Test 6.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P25.** What is the default port number for DNS?DNS

**(A)** 80
**(B)** 21
**(C)** 53
**(D)** 110

[Source: CN Topic Test 6.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ auto-ingested — answer needs verification

**P26.** In IPv6, which of the following is true?IPv6

**(A)** Uses 32-bit address
**(B)** Includes checksum in header
**(C)** Supports anycast addressing
**(D)** Single amplification

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / IPv6 Addressing]

⚠ auto-ingested — answer needs verification

**P27.** Which address type is unique in the network and globally routable in IPv6?IPv6

**(A)** Link-Local
**(B)** Multicast
**(C)** Unique local
**(D)** Global unicast

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / IPv6 Addressing]

⚠ auto-ingested — answer needs verification

**P28.** What is the maximum number of hosts in a /24 subnet? 24

**(A)** 128
**(B)** 256
**(C)** 254
**(D)** 512

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P29.** What is the typical range of ephemeral ports? Ephemeral

**(A)** 0-1023
**(B)** 1024-49151
**(C)** 49152-65535
**(D)** None of the these

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ auto-ingested — answer needs verification

**P30.** subnet mask of 255.255.255.248 allows how many usable hosts?255.255.255.248 usable

**(A)** 2
**(B)** 6
**(C)** 14
**(D)** 30

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P31.** What is the function of IGMP in networking? IGMP

**(A)** Manages IP address allocation
**(B)** Routes datagrams
**(C)** Manages multicast group membership
**(D)** Handles device discovery

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / MAC Address]

⚠ auto-ingested — answer needs verification

**P32.** What is the maximum number of IP addresses in Class C?Class C IP

**(A)** 254
**(B)** 256
**(C)** 1024
**(D)** 65536

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P33.** Which type of DNS record maps IP to hostname? - DNS IP

**(A)** MX
**(B)** A
**(C)** PTR
**(D)** CNAME

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ auto-ingested — answer needs verification

**P34.** In subnetting, reducing the number of host bits does what

**(A)** Increase broadcast domain
**(B)** Decrease number of subnets
**(C)** Increase number of subnets
**(D)** all Increase number of hosts

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P35.** What type of address is FF:FF:FF:FF:FF:FF? FF:FF:FF:FF:F F:FF

**(A)** Unicast
**(B)** Multicast
**(C)** Boardcast
**(D)** Loopback

[Source: CN Topic Test 5.pdf | auto-classified: Topic 3 / MAC Address]

⚠ auto-ingested — answer needs verification

**P36.** MAC address is also called _____. _____

**(A)** Physical address
**(B)** Logical address
**(C)** Source address
**(D)** Destination address

[Source: CN Topic Test-01.pdf | auto-classified: Topic 3 / MAC Address]

⚠ auto-ingested — answer needs verification

**P37.** What is the main purpose of a Domain Name System (DNS) server? (DNS)

**(A)** To allocate IP addresses dynamically
**(B)** To convert domain names into IP addresses
**(C)** To route traffic within a LAN
**(D)** To manage firewall settings

[Source: CN Topic Test-02.pdf | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ auto-ingested — answer needs verification

**P38.** There is a need to create a network that has 5 subnets, each with at least 16 hosts. which one is used as classful subnet mask? 16

**(A)** 255.255.255.192
**(B)** 255.255.255.248
**(C)** 255.255.255.240
**(D)** 255.255.255.224

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P39.** Which of the following is a class B host address
**(A)** 230.0.0.0
**(B)** 130.4.5.6
**(C)** 230.7.6.5
**(D)** 30.4.5.6
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P40.** A subnet mask in class C can have _____ 1's with the remaining bits O'sb C _____ 1
**(A)** 10
**(B)** 24
**(C)** 12
**(D)** 7
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P41.** IPv6 is developed by IPv6
**(A)** IETF
**(B)** ANSI
**(C)** ISO
**(D)** IEEE
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv6 Addressing]

⚠ auto-ingested — answer needs verification

**P42.** In the IPv4 addressing format, the number of networks allowed under Class C addresses is IPv4 , C
**(A)** 220
**(B)** 224
**(C)** 214
**(D)** 221
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P43.** Which of the following statement/s is/are true? (i) Windows XP supports both peer- peer and client-server networks. (ii) Windows XP implements Transport protocols as drivers that can be loaded and uploaded from the system dynamically. (i) (ii) Choose the correct answer from the code given below:
**(A)** Both (i) and (ii)
**(B)** Neither (i) nor (ii) (ii) only (i) only (ii) 11. The four byte IP Address consists of (a) Neither network nor Host Address (b) Network Address
**(C)** Both Network and Host Address
**(D)** Host Address
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P44.** The dotted-decimal notation of the following IPV4 address in binary notation is IPV4
**(A)** 111.56.45.239
**(B)** 129.11.10.238
**(C)** 129.11.11.239
**(D)** 111.56.11.239
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P45.** What is the maximum number of hosts that can be assigned IP addresses on a /24 network? 24 IP

**(A)** 254
**(B)** 256
**(C)** 512
**(D)** 1024

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P46.** In a class B subnet, we know the IP address of one host and the mask as given below: IP address: 125.134.112.66 Mask: 255.255.224.0 What is the first address (Network address)

**(A)** 125.134.96.0
**(B)** 125.134.112.0
**(C)** 125.134.112.66
**(D)** 125.134.0.0

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P47.** In CIDR notation, what does the /16 represent? CIDR

**(A)** 16 host bits
**(B)** 16 network bits
**(C)** 16 bytes
**(D)** 16 subnets

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P48.** Which IP address class provides the largest number of hosts per network

**(A)** Class A
**(B)** Class B
**(C)** Class C
**(D)** Class D

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P49.** If the number of networks and number of hosts in class B are 2m, (2n −2) respectively. Then the relation between m,n is: 2m (2n −2) m,n :

**(A)** 3m=2n
**(B)** 7m=8n
**(C)** 8m=7n
**(D)** 2m=3n

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P50.** The address 255.255.255.255 is used for:

**(A)** A public IP
**(B)** A default gateway
**(C)** A network ID
**(D)** Broadcast to all nodes

[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P51.** What is the range of addresses for APIPA (Automatic Private IP Addressing)? APIPA ( )
**(A)** 169.254.0.1 to 169.254.255.254
**(B)** 10.0.0.0 to 10.255.255.255
**(C)** 192.168.0.0 to 192.168.255.255
**(D)** 172.16.0.0 to 172.31.255.255
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P52.** Which part of the IP address 172.16.5.4/24 identifies the network? 172.16.5.4/24
**(A)** 172
**(B)** 172.16
**(C)** 172.16.5
**(D)** 172.16.5.4
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P53.** The first octet of a Class B IP address falls between:
**(A)** 0-127
**(B)** 128-191
**(C)** 192-223
**(D)** 224-239
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P54.** What is the purpose of subnetting
**(A)** To increase the number of hosts per network
**(B)** To divide a network into smaller networks
**(C)** To reduce the number of IP addresses needed
**(D)** To merge multiple networks
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P55.** What is the valid host range for the subnet 192.168.1.0/25? 192.168.1.0/25
**(A)** 192.168.1.0 - 192.168.1.255
**(B)** 192.168.1.1 - 192.168.1.126
**(C)** 192.168.1.1 - 192.168.1.254
**(D)** 192.168.1.0 - 192.168.1.127
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P56.** In a subnet mask, the number of 1's represents:
**(A)** Host bits
**(B)** Network bits
**(C)** Broadcast bits
**(D)** Subnet bits The IP address 224.0.0.1 belongs to: IP address 224.0.0.1 : (a) Private IP address (b) Public IP address (c) Multicast address (d) Broadcast address
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P57.** How many usable hosts are there in a /30 subnet? 30
**(A)** 2
**(B)** 4
**(C)** 6
**(D)** 14 40 What does NAT (Network Address Translation) primarily help with? NAT ( ) (a) Speeding up IP packets (b) Connecting two networks (c) IP address conservation (d) Encrypting IP traffic
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P58.** What is the purpose of the "reserved" IP address 0.0.0.0? reserved IP address 0.0.0.0
**(A)** Default gateway
**(B)** Used for DHCP servers
**(C)** Default route or "any network"
**(D)** Multicast address
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P59.** Which header field is removed when an IP packet moves across a router? IP
**(A)** Source IP address
**(B)** Destination IP address
**(C)** MAC (Media Access Control) address
**(D)** Subnet mask
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P60.** The subnet mask 255.255.255.128 provides how many subnets for a Class C network? 255.255.255.128
**(A)** 1
**(B)** 2
**(C)** 4
**(D)** 8
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P61.** Which IP address type is used to communicate with all devices on a subnet
**(A)** Unicast
**(B)** Broadcast
**(C)** Multicast
**(D)** Anycast
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / MAC Address]

⚠ auto-ingested — answer needs verification

**P62.** The address of a class B host o be split into subnets with a 3 bit subnet number. What is the maximum number of subnets and maximum number of hosts in each subnet
**(A)** 8 subnets and 262141 hosts
**(B)** 6 subnets and 262141 hosts
**(C)** 6 subnets and 1022 hosts
**(D)** None of the Above
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P63.** The special IP address used to test network interface (NIC) hardware is: ( )
**(A)** 127.0.0.1
**(B)** 0.0.0.0
**(C)** 255.255.255.255
**(D)** 169.254.0.1 The process of hiding internal IP addresses from external networks is called: IP (a) Port forwarding (b) NAT (c) Tunneling (d) VPN
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P64.** Which class of IP addresses supports 65,534 hosts per network? 65,534
**(A)** Class A
**(B)** Class B
**(C)** Class C
**(D)** Class D
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P65.** Which IP address is reserved for future use and experimental purposes
**(A)** 127.0.0.1
**(B)** 224.0.0.1
**(C)** 240.0.0.0
**(D)** 192.168.0.1
[Source: CN Topic Test-03.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P66.** An ethernet destination address 0A-01-12-03-04-05 is: 0A-01-12-03-04 05 :
**(A)** Unicast address
**(B)** Multicast address
**(C)** Broadcast address
**(D)** All of the options
[Source: CN Topic Test-04.pdf | auto-classified: Topic 3 / MAC Address]

⚠ auto-ingested — answer needs verification

**P67.** The IP address is used by hosts when they are being booted
**(A)** 0.0.0.0
**(B)** 1.0.0.0
**(C)** 1.1.1.1
**(D)** 255.255.255.255
[Source: CN Topic Test-04.pdf | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P68.** The address used in the Data Link Layer is called:
**(A)** Logical address
**(B)** Port address
**(C)** MAC address
**(D)** IP address
[Source: CN Topic Test-04.pdf | auto-classified: Topic 3 / MAC Address]

⚠ auto-ingested — answer needs verification

**P69.** What does /24 denote in CIDR?
**(A)** 24 bits are used for host portion
**(B)** 24 bits are used for network portion
**(C)** 24 subnets are created
**(D)** 24 addresses are available in the subnet
[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P70.** IP address 172.16.123.213 is found to be residing on a sub-network with the following subnet mask: 255.255.255.240. What is the network address of this subnet?
**(A)** 172.16.123.240
**(B)** 172.16.123.208
**(C)** 172.16.123.213
**(D)** 172.16.123.245
[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P71.** The size of the 'total length field' of IPv4 datagram is -
**(A)** 16 bits
**(B)** 32 bits
**(C)** 8 bits
**(D)** 4 bits
[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P72.** What is the role of DNS? It is best described as -
**(A)** maps a domain name to an IP address
**(B)** translate a host's private IP address to a public IP address
**(C)** translate a host's IPv6 address to an IPv4 address
**(D)** maps a MAC address to an IP address
[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P73.** A Reverse Lookup in DNS means a lookup to ..... using ..... records.
**(A)** identify mail servers, SOA
**(B)** map IP address to domain name, PTR
**(C)** map domain names to IP address, A
**(D)** determine the authoritative name servers, CNAME
[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P74.** An organization has a Class B network and wishes to form subnets for 64 departments. The subnet mask would be -
**(A)** 255.255.128.0
**(B)** 255.255.0.0
**(C)** 255.255.64.0
**(D)** 255.255.252.0
[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P75.** If a Class B network on the internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?
**(A)** 2046
**(B)** 1022
**(C)** 1023
**(D)** 1024
[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ auto-ingested — answer needs verification

**P76.** The Address Resolution Protocol (ARP) is used for -
**(A)** finding the IP address that corresponds to a MAC address
**(B)** finding the MAC address that corresponds to an IP address
**(C)** finding the IP address from the DNS
**(D)** finding the IP address of the default gateway
[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ auto-ingested — answer needs verification

**P77.** The size of MAC address (in bits) is -
**(A)** 42
**(B)** 48
**(C)** 24
**(D)** 36
[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 3 / MAC Address]

⚠ auto-ingested — answer needs verification

**P78.** URL belongs to which layer? URL ?**
**(A)** Presentation
**(B)** Application
**(C)** Transport
**(D)** Network
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 1 / General Network Basics]

⚠ auto-ingested — answer needs verification

**P79.** S1: Every Computer or an Internet device has a unique MAC address. S2: Every Computer or an Internet device has a unique IP address. Consider the following statement given above ? S1: MAC S2: IP
**(A)** Statement s1 true and s2 false
**(B)** Both statement s1,s2 false
**(C)** Staement s1 false and s2 true
**(D)** Both statement s1,s2 true
[Source: CN Topic Test-01.pdf | auto-classified: Topic 1 / Types of Networks]

⚠ auto-ingested — answer needs verification

**P80.** A host is connected to a Department network which is part of a University network. The university network, in turn is part of the Internet. The largest network in which the Ethernet address of the host in unique is

**(A)** The subnet to which the host belongs
**(B)** The department network
**(C)** The university network
**(D)** the Internet

[Source: CN Topic Test-02.pdf | auto-classified: Topic 1 / Types of Networks]

⚠ auto-ingested — answer needs verification

**P81.** IP address can be used in WAN IP WAN

**(A)** 256.0.0.1
**(B)** 172.16.0.10
**(C)** 15.1.5.6
**(D)** 127.256.0.118.

[Source: CN Topic Test-03.pdf | auto-classified: Topic 1 / Types of Networks]

⚠ auto-ingested — answer needs verification

**P82.** Which organization is responsible for managing IP address allocation globally

**(A)** IEEE
**(B)** ICANN
**(C)** IETF
**(D)** ISO

[Source: CN Topic Test-03.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

**P83.** The size of an IP address in IPv6 is:

**(A)** 4 bytes
**(B)** 128 bits
**(C)** 8 bytes
**(D)** 100 bits

[Source: Sanfoundry CN MCQs — IPv6 | auto-classified: Topic 3 / IPv4 Addressing]

⚠ Site key.

**P84.** IPv6 does NOT use which type of address?

**(A)** Broadcast
**(B)** Multicast
**(C)** Anycast
**(D)** None of the mentioned

[Source: Sanfoundry CN MCQs — IPv6 | auto-classified: Topic 3 / IPv6 Addressing]

⚠ Site key: broadcast eliminated in IPv6.

**P85.** Which features are present in IPv4 but not in IPv6?

**(A)** Fragmentation by routers
**(B)** Header checksum
**(C)** Options
**(D)** All of the mentioned

[Source: Sanfoundry CN MCQs — IPv6 | auto-classified: Topic 3 / IPv4 Addressing]

⚠ Site key.

| | |
|---|---|
| **P86.** Two IPv6 nodes are connected through intervening IPv4 routers. The best solution to interoperate is:<br>**(A)** Use the dual-stack approach<br>**(B)** Tunneling<br>**(C)** No solution<br>**(D)** Replace the system<br>[Source: Sanfoundry CN MCQs — IPv6 \| auto-classified: Topic 3 / IPv4 Addressing] | ⚠ Site key: IPv4 routers form a tunnel. |
| **P87.** MAC address is of size:<br>**(A)** 24 bits<br>**(B)** 36 bits<br>**(C)** 42 bits<br>**(D)** 48 bits<br>[Source: Sanfoundry CN MCQs — Data Link \| auto-classified: Topic 3 / MAC Address] | ⚠ Site key + Forouzan. |
| **P88.** The size of a MAC address is:<br>**(A)** 16 bits<br>**(B)** 32 bits<br>**(C)** 48 bits<br>**(D)** 64 bits<br>[Source: InstrumentationTools 150 CN MCQs \| auto-classified: Topic 3 / MAC Address] | ⚠ Verified (dup of Sanfoundry wording, kept as separate bank item). |
| **P89.** Which addresses are used in an internet employing the TCP/IP protocols?<br>**(A)** Physical address and logical address<br>**(B)** Port address<br>**(C)** Specific address<br>**(D)** All of these<br>[Source: LMT MCQ bank (Computer Communication Networks) \| auto-classified: Topic 3 / MAC Address] | ⚠ Site key. Matches Forouzan Ch1: physical/IP/port/ specific address levels. |
| **P90.** Which addressing scheme is used by IPv6?<br>**(A)** 32-bit<br>**(B)** 128-bit<br>**(C)** 64-bit<br>**(D)** 16-bit<br>[Source: 30 important CN MCQs (Medium blog bank) \| auto-classified: Topic 3 / IPv6 Addressing] | ⚠ Site key. |
| **P91.** What is the primary purpose of DNS (Domain Name System)?<br>**(A)** Assigning IP addresses to devices<br>**(B)** Resolving domain names to IP addresses<br>**(C)** Encrypting network communication<br>**(D)** Managing network resources<br>[Source: 30 important CN MCQs (Medium blog bank) \| auto-classified: Topic 3 / Port Numbers & DNS/URL] | ⚠ Site key. |

**P92.** Which subnet mask is associated with the Class B IP address range?
**(A)** 255.255.255.0
**(B)** 255.0.0.0
**(C)** 255.255.0.0
**(D)** 255.255.255.255
[Source: 30 important CN MCQs (Medium blog bank) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ Site key.

**P93.** What is the function of a DNS resolver in the DNS hierarchy?
**(A)** Translating domain names to IP addresses
**(B)** Managing top-level domain servers
**(C)** Resolving IP addresses to domain names
**(D)** Authenticating domain registrations
[Source: 30 important CN MCQs (Medium blog bank) | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ Site key.

**P94.** Which of the following is a valid IPv4 address?
**(A)** 192.256.1.1
**(B)** 172.31.0.1
**(C)** 300.100.50.25
**(D)** 500.10.5.2
[Source: 30 important CN MCQs (Medium blog bank) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ Site key.

**P95.** What does the term 'subnetting' refer to in networking?
**(A)** Dividing a large network into smaller logical networks
**(B)** Allocating IP addresses to devices
**(C)** Filtering network traffic
**(D)** Assigning MAC addresses
[Source: 30 important CN MCQs (Medium blog bank) | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ Site key.

**P96.** When IPv4 addressing is manually configured on a web server, which property identifies the network and host portions of the address?
**(A)** DNS server address
**(B)** Subnet mask
**(C)** Default gateway
**(D)** DHCP server address
[Source: IP Addressing quiz (Studocu bank, with keys) | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ Site key.

**P97.** What is the purpose of NAT (Network Address Translation) in IPv4?
**(A)** To map private IP addresses to a single public IP address
**(B)** To increase the security of IP addresses
**(C)** To facilitate communication between different subnets
**(D)** All of the above
[Source: IP Addressing quiz (Studocu bank, with keys) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ Site key.

**P98.** What is the size of each octet in an IPv4 address?
**(A)** 4 bits
**(B)** 8 bits
**(C)** 16 bits
**(D)** 32 bits
[Source: IP Addressing quiz (Studocu bank, with keys) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ Site key.

**P99.** What is the maximum number of unique IPv4 addresses?
**(A)** 2^16
**(B)** 2^24
**(C)** 2^32
**(D)** 2^64
[Source: IP Addressing quiz (Studocu bank, with keys) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ Site key.

**P100.** What is the address size of IPv6?
**(A)** 32 bit
**(B)** 64 bit
**(C)** 128 bit
**(D)** 256 bit
[Source: Networking MCQ bank (MQuestions blog) | auto-classified: Topic 3 / IPv6 Addressing]

⚠ Verified.

**P101.** DNS is the abbreviation of:
**(A)** Dynamic Name System
**(B)** Dynamic Network System
**(C)** Domain Name System
**(D)** Domain Network Service
[Source: Networking MCQ bank (MQuestions blog) | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ Verified.

**P102.** An IPv4 address is of size:
**(A)** 8 bit
**(B)** 16 bit
**(C)** 32 bit
**(D)** 64 bit
[Source: Networking MCQ bank (MQuestions blog) | auto-classified: Topic 3 / IPv4 Addressing]

⚠ Verified.

**P103.** 1. Which of the following DNS Resource Records should be queried to retrieve the IPv6 address associated to a name ?
**(A)** The `AAAA` record.
**(B)** The `A` record.
**(C)** The `NS` record.
**(D)** The `MX` record.
[Source: cnp3 MCQ (Bonaventure CN textbook), DNS1 | auto-classified: Topic 3 / IPv6 Addressing]

⚠ auto-ingested — answer needs verification

**P104.** 2. You use the DNS to retrieve the `IPv4` address associated to the name `cnp3book.info.ucl.ac.be`. Assuming that you need to start your query from the root of the domain name system, which of the following DNS record will you *never* query to obtain this information.
**(A)** The `AAAA` record.
**(B)** The `A` record.
**(C)** The `NS` record.
**(D)** The `MX` record.
[Source: cnp3 MCQ (Bonaventure CN textbook), DNS2 | auto-classified: Topic 3 / Port Numbers & DNS/URL]

⚠ auto-ingested — answer needs verification

**P105.** Among the following binary representations, which is the one that corresponds to address ``2001:DB8:1341:FC81::1`` (the first line shows the higher order 64 bits starting from the highest order bits, the second the low order 64 bits) ?
**(A)** .. code-block:: console
**(B)** .. code-block:: console
**(C)** .. code-block:: console
**(D)** .. code-block:: console
[Source: cnp3 MCQ (Bonaventure CN textbook), ipv6addr | auto-classified: Topic 3 / IPv6 Addressing]

⚠ auto-ingested — answer needs verification

**P106.** Prafocd 71. ICMP protocol is a . layer protocol. (1) Data link (2) Application (3) Transport Classless.Tnter Domain (5) CIDR 72. What does /24 denote in CIDR?
**(A)** (1)
**(B)** (2)24 addresses are available in the subnet
**(C)** 24bits areused fornetworkportion
**(D)** (4) 24 bits are used for host portion
[Source: Random Test- Rajasthan PYQ.pdf (OCR), p22 | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

**P107.** 3. given an IP address 156.233.42.56 with subnet mask of 7 bits. How many hosts and subnets are possible ?
**(A)** 126 hosts, 510 subnets 42.SG
**(B)** 128 hosts , 512 subnets 156. 233
**(C)** 510 hosts 126 subnets
**(D)** 512 hosts, 128 subnets UasB Host 29-2 2 -2 (26 5r
[Source: Unit Test -2 Solutions.pdf (OCR), p5 | auto-classified: Topic 3 / Subnetting & CIDR]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

**P108.** 28. Lightweight Directory Access Protocol is used for
**(A)** Routing the packets (byAuthentication
**(B)** Obtaining IP address
**(C)** Domain name resolving
[Source: Unit Test-4 Solution.pdf (OCR), p29 | auto-classified: Topic 3 / IPv4 Addressing]

⚠ OCR-parse staging: verify options/answer at curation (see src page).